

# Jiajun Zhang

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## EDUCATION

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### Bachelor of Science in Honors Applied Mathematics

McGill University

2022-Present

Graduation Time: June 2026

## RESEARCH INTERESTS

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Mathematical Statistics, Biostatistics, and their connections with Machine Learning, Deep Learning and Casual Inference.

## MATHEMATICAL COURSEWORK

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Probability and Mathematical Statistics; Regression and ANOVA; Differential Equations and Dynamical Systems; Real Analysis and Abstract Algebra; Calculus and Linear Algebra

## RESEARCH EXPERIENCE

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### Honors Research Project

McGill University

2025.2 - 2025.8

- Research Title: Analyzing Incident and Prevalent Cohort Survival Data
- Research Supervisor: Professor Masoud Asgharian
- Written Report: *An Introduction to Survival Analysis*

This research provides a review of techniques in analyzing right-censored, left-truncated cohort survival data. Approaches include parametric model using likelihood inference; non-parametric model using Kaplan-Meier, Nelson-Aalen and multi-state Johansen-Aalen estimators and their properties; Cox's semi-parametric hazard regression model; Non-parametric regression and hypothesis testing models.

## UNDERGRADUATE PROJECTS

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### Dynamical Systems

2025.3 - 2025.4

- Project Report: Introduction to Circular Restricted Three-Body Problem
- Project Supervisor: Professor Antony R. Humphries

This project studied the dynamics of a circular restricted three body problem. Motivated from the closed solution of a two body problem, an additionally massless object is added into the system and derived the set of Lagrange points of the third body as well as their corresponding orbits and stability.

## TEACHING AND MENTORING EXPERIENCE

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### Peer Tutoring

McGill University

2023.9 - Present

- Provided individualized help to mentee in mathematics via advice and tutoring.

### Undergraduate Mathematics Helpdesk Mentor

McGill University

2024.9 - Present

- Offer open office hours each week to guide students in need and help them with their mathematics coursework.

## Undergraduate Course Assistant

McGill University

- Grading student's homework and provide feedback to professors.
  - Course Assistant for MATH 254: Honors Analysis 1 2025.9 - Present
  - Course Assistant for MATH 248: Honors Vector Calculus 2024.9 - 2024.12
  - Course Assistant for MATH 223: Linear Algebra 2023.9 - 2024.4

## HONORS AND AWARDS

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### Tomlinson Undergraduate Award

2025.3 and 2024.10

McGill University

- Awarded to undergraduate course assistants.

## VOLUNTEERING EXPERIENCE

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### Math & Stats Buddy Mentorship Program

2025.9 - Present

McGill University

- This program matches upper year undergraduate students from math department with freshman students, where mentor will provide advice and help for mentee.

### McGill Robotics Club

2023.9 - 2024.4

McGill University

- Duty: Working with Drone Group as a software developer on drone's vision system.

## EXTRACURRICULAR PROJECTS

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### Math Resources Website

2024 - Present

- Website link: [Math Resources](#)
- Developed a website to provide free math resources for undergraduate students and help them learn.

### Video Channel

2023 - Present

- Provide entertainment and math-themed educational videos for around 7,200 subscribers.
- Account page: [Video Channel](#)

## SKILLS

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- **Technical Skills:** LaTeX, Github, R, Python, Java

## LANGUAGES

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Mandarin (Native), English (Proficient), French (Intermediate)